task #2

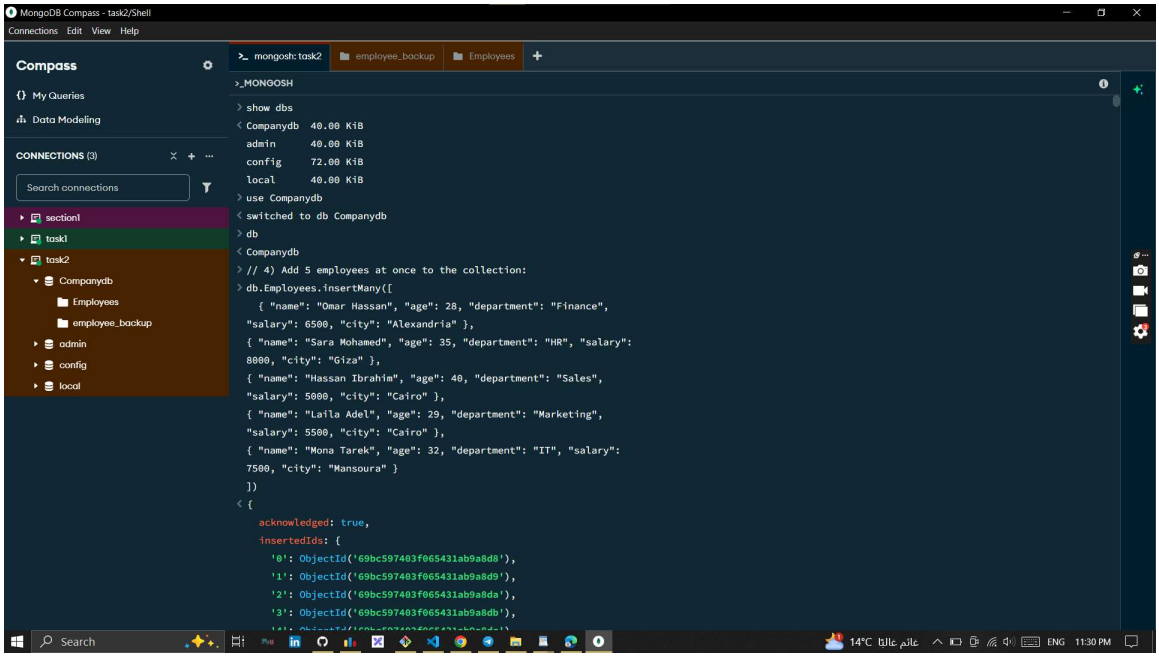
name : asmaa ahmed maryah

id : 811634546

aggregation piplines

---

1) Create a new database called Companydb and add a Collection set as Employees.



2) Import Employees_Large.json data to database.

3) Add a new employee to the collection with following data:

- "name": "Ahmed Ali"
- "age": 30
- "department": "IT"
- "salary": 7000
- "city": "Cairo"

4) Add 5 employees at once to the collection:

{ "name": "Omar Hassan", "age": 28, "department": "Finance",

"salary": 6500, "city": "Alexandria" },

{ "name": "Sara Mohamed", "age": 35, "department": "HR", "salary":

8000, "city": "Giza" },

{ "name": "Hassan Ibrahim", "age": 40, "department": "Sales",

"salary": 5000, "city": "Cairo" },

{ "name": "Laila Adel", "age": 29, "department": "Marketing",

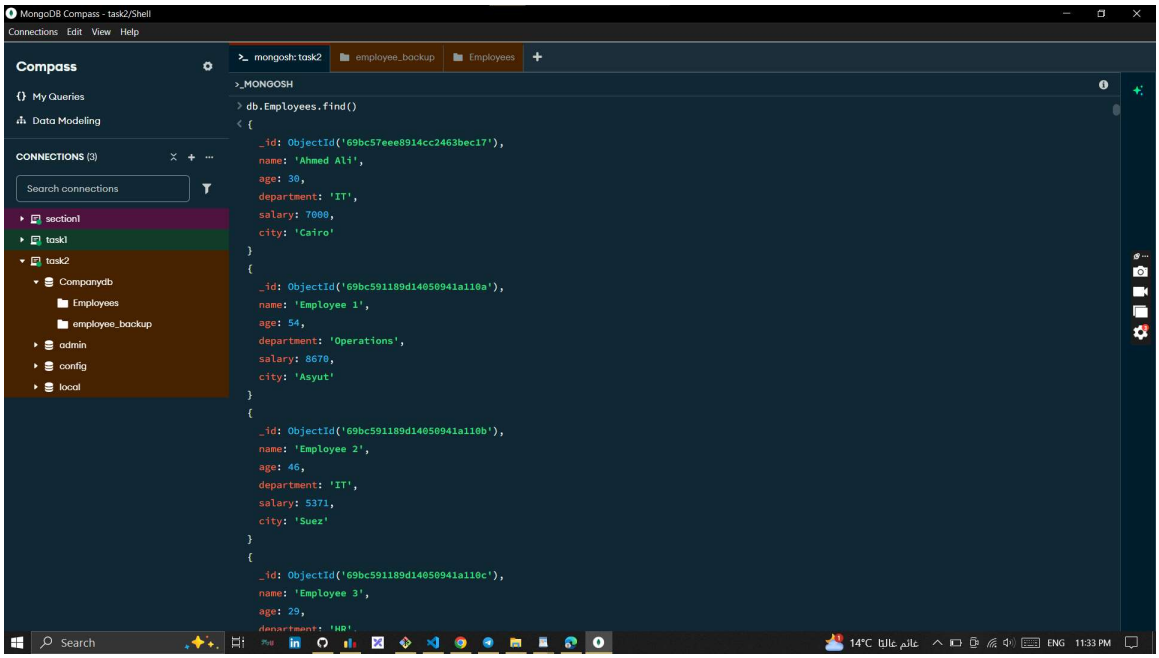
"salary": 5500, "city": "Cairo" },

{ "name": "Mona Tarek", "age": 32, "department": "IT", "salary":

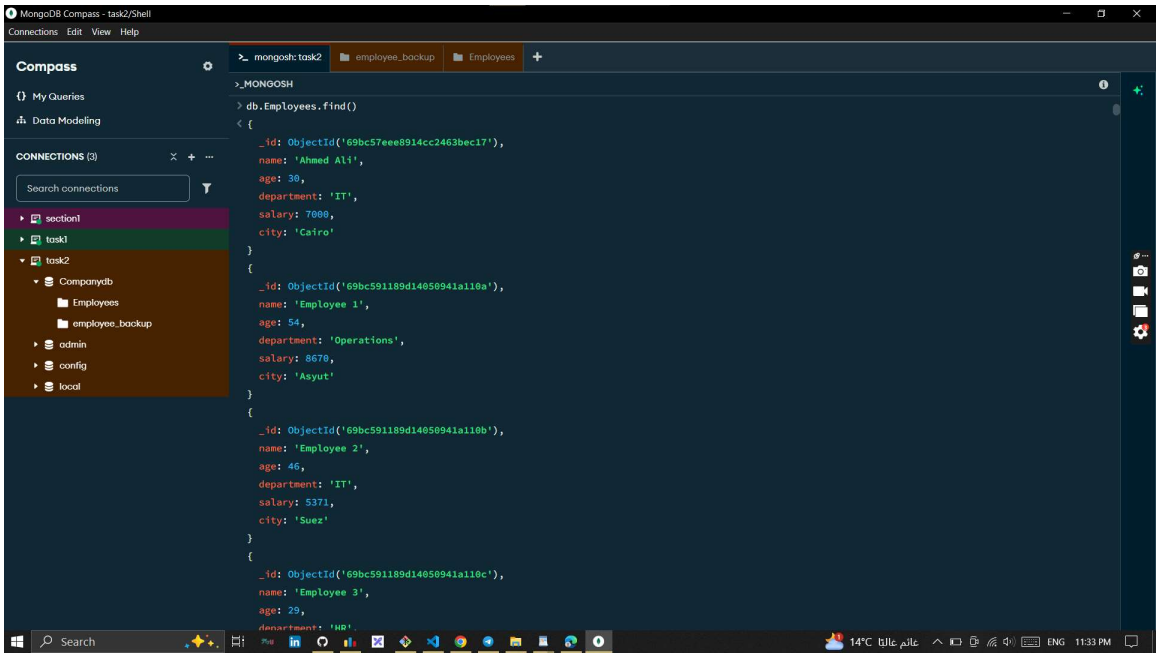
7500, "city": "Mansoura" }

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69bc597403f065431ab9a8d8'),
    '1': ObjectId('69bc597403f065431ab9a8d9'),
    '2': ObjectId('69bc597403f065431ab9a8da'),
    '3': ObjectId('69bc597403f065431ab9a8db'),
    '4': ObjectId('69bc597403f065431ab9a8dc')
  }
}
```

5) Show all employees in the collection.



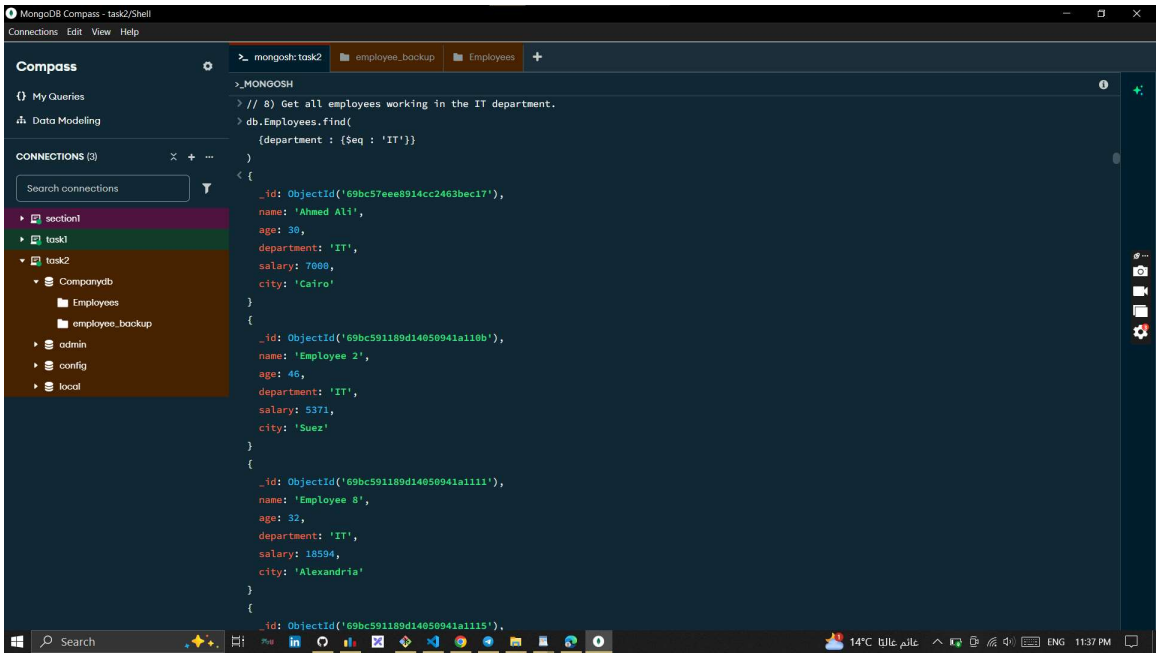
6) Get only 5 employees from the collection.

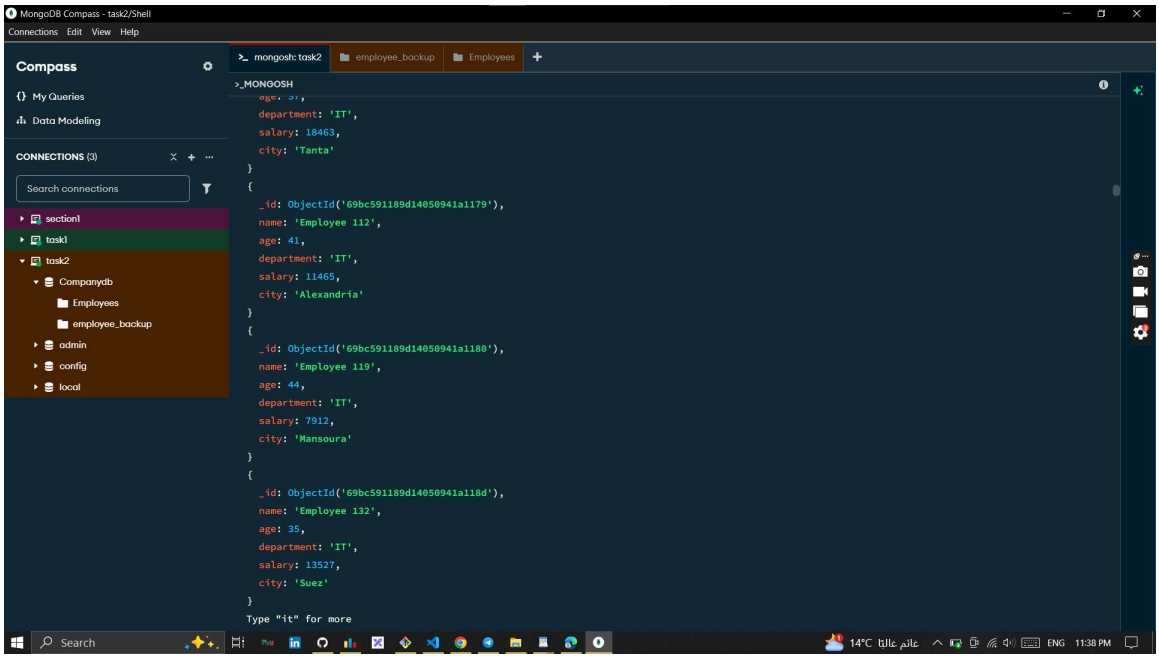


7) Find a specific employee by name.

```
> // Find a specific employee by name.
> db.Employees.find(
    {name : {$eq : "Ahmed Ali" }}
  )
< {
    _id: ObjectId('69bc57eee8914cc2463bec17'),
    name: 'Ahmed Ali',
    age: 30,
    department: 'IT',
    salary: 7000,
    city: 'Cairo'
  }
```

8) Get all employees working in the IT department.





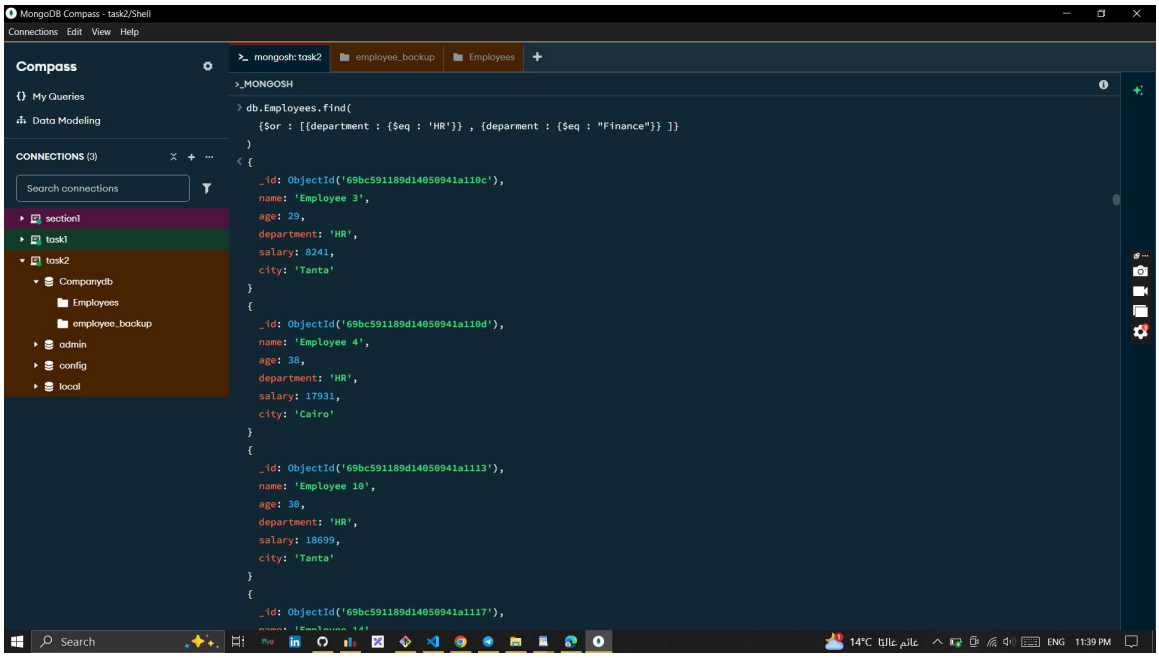
9) Get all employees whose salaries are greater than 5000.

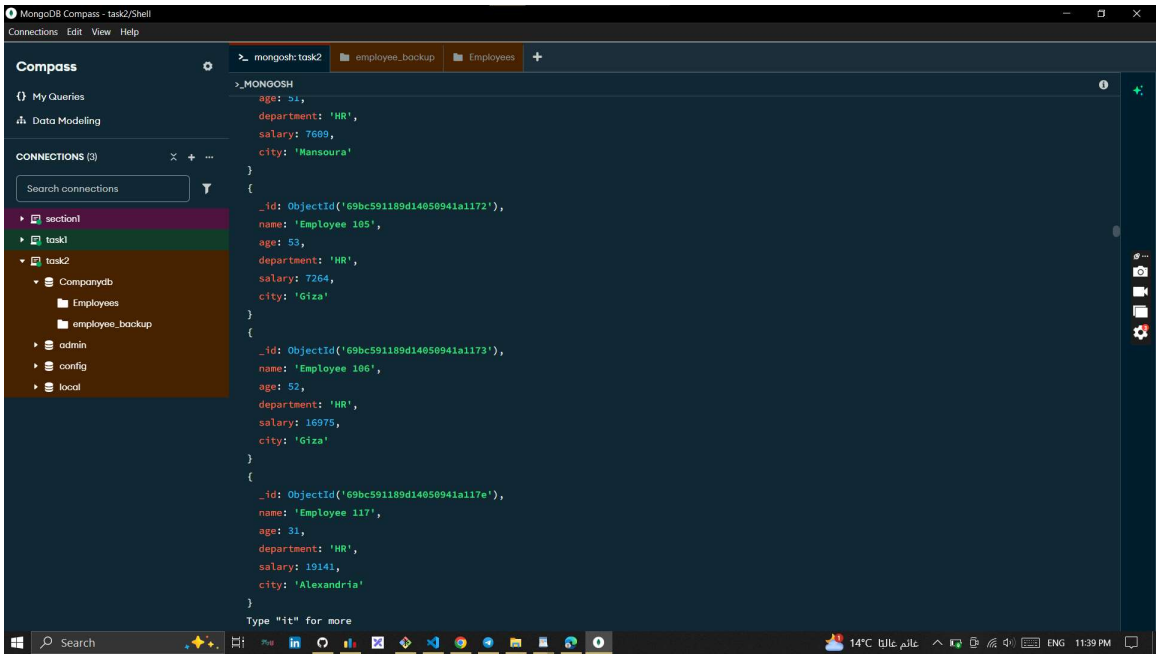
```
> db.Employees.find({salary : {$gt : 5000} })
< {
    _id: ObjectId('69bc57eee8914cc2463bec17'),
    name: 'Ahmed Ali',
    age: 30,
    department: 'IT',
    salary: 70000,
    city: 'Cairo'
  }
  {
    _id: ObjectId('69bc591189d14050941a110a'),
    name: 'Employee 1',
    age: 54,
    department: 'Operations',
    salary: 86700,
    city: 'Asyut'
  }
  {
    _id: ObjectId('69bc591189d14050941a110b'),
    name: 'Employee 2',
    age: 46,
    department: 'IT',
    salary: 53710,
    city: 'Suez'
  }
  {
    _id: ObjectId('69bc591189d14050941a110c'),
    name: 'Employee 3',
    age: 29,
    department: 'HR',
```

```
{
  _id: ObjectId('69bc591189d14050941a110c'),
  name: 'Employee 3',
  age: 29,
  department: 'HR',
  salary: 82410,
  city: 'Tanta'
}
{
  _id: ObjectId('69bc591189d14050941a110d'),
  name: 'Employee 4',
  age: 38,
  department: 'HR',
  salary: 179310,
  city: 'Cairo'
}
{
  _id: ObjectId('69bc591189d14050941a110e'),
  name: 'Employee 5',
  age: 44,
  department: 'Operations',
  salary: 58480,
  city: 'Giza'
}
{
  _id: ObjectId('69bc591189d14050941a110f'),
  name: 'Employee 6',
  age: 35,
  department: 'Operations',
  salary: 82220,
```

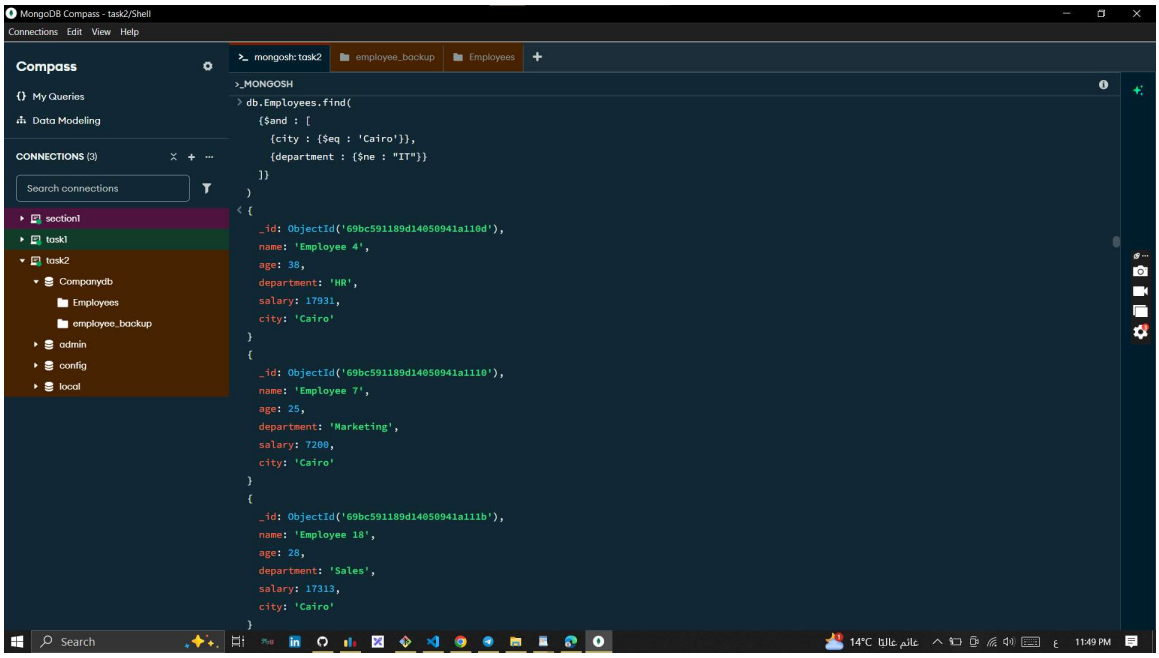
```
  age: 22,
  department: 'Operations',
  salary: 171700,
  city: 'Alexandria'
}
{
  _id: ObjectId('69bc591189d14050941a111a'),
  name: 'Employee 17',
  age: 59,
  department: 'Sales',
  salary: 163590,
  city: 'Asyut'
}
{
  _id: ObjectId('69bc591189d14050941a111b'),
  name: 'Employee 18',
  age: 28,
  department: 'Sales',
  salary: 173130,
  city: 'Cairo'
}
{
  _id: ObjectId('69bc591189d14050941a111c'),
  name: 'Employee 19',
  age: 47,
  department: 'Sales',
  salary: 159020,
  city: 'Tanta'
}
Type "it" for more
```

10) Get all employees working in the "HR" or "Finance" department.





11) Get all employees who live in "Cairo" and do not work in the "IT" department.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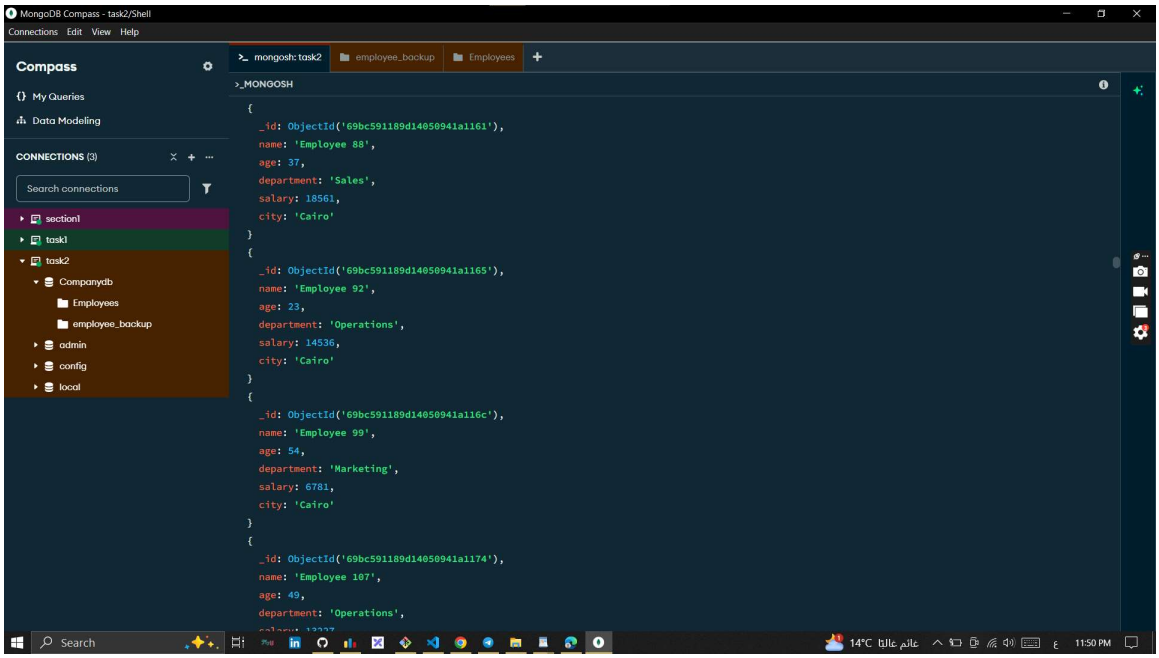
>_MONGOSH
> db.Employees.find(
{$and : [
{city : {$eq : 'Cairo'}},
{department : {$ne : "IT"}}
]}
)
< {
_id: ObjectId('69bc591189d14050941a110d'),
name: 'Employee 4',
age: 38,
department: 'HR',
salary: 17931,
city: 'Cairo'
}
{
_id: ObjectId('69bc591189d14050941a1110'),
name: 'Employee 7',
age: 25,
department: 'Marketing',
salary: 7200,
city: 'Cairo'
}
{
_id: ObjectId('69bc591189d14050941a111b'),
name: 'Employee 18',
age: 28,
department: 'Sales',
salary: 17313,
city: 'Cairo'
}
Search
14°C
11:49 PM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{
_id: ObjectId('69bc591189d14050941a1161'),
name: 'Employee 88',
age: 37,
department: 'Sales',
salary: 18561,
city: 'Cairo'
}
{
_id: ObjectId('69bc591189d14050941a1165'),
name: 'Employee 92',
age: 23,
department: 'Operations',
salary: 14536,
city: 'Cairo'
}
{
_id: ObjectId('69bc591189d14050941a116c'),
name: 'Employee 99',
age: 54,
department: 'Marketing',
salary: 6781,
city: 'Cairo'
}
{
_id: ObjectId('69bc591189d14050941a1174'),
name: 'Employee 107',
age: 49,
department: 'Operations',
Search
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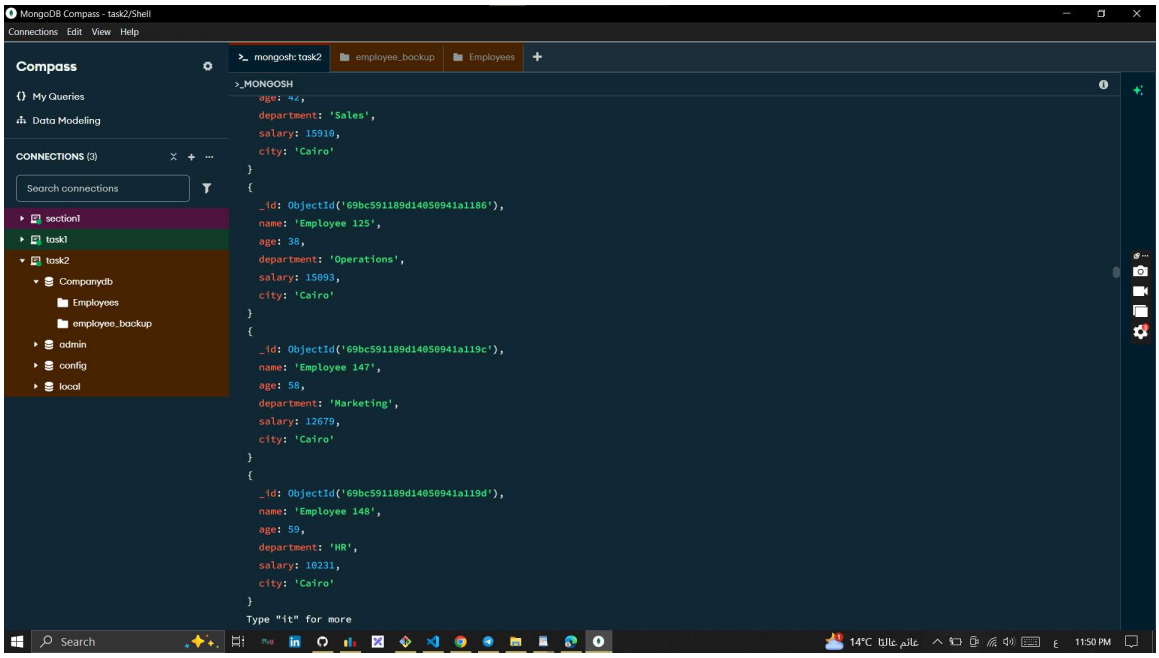
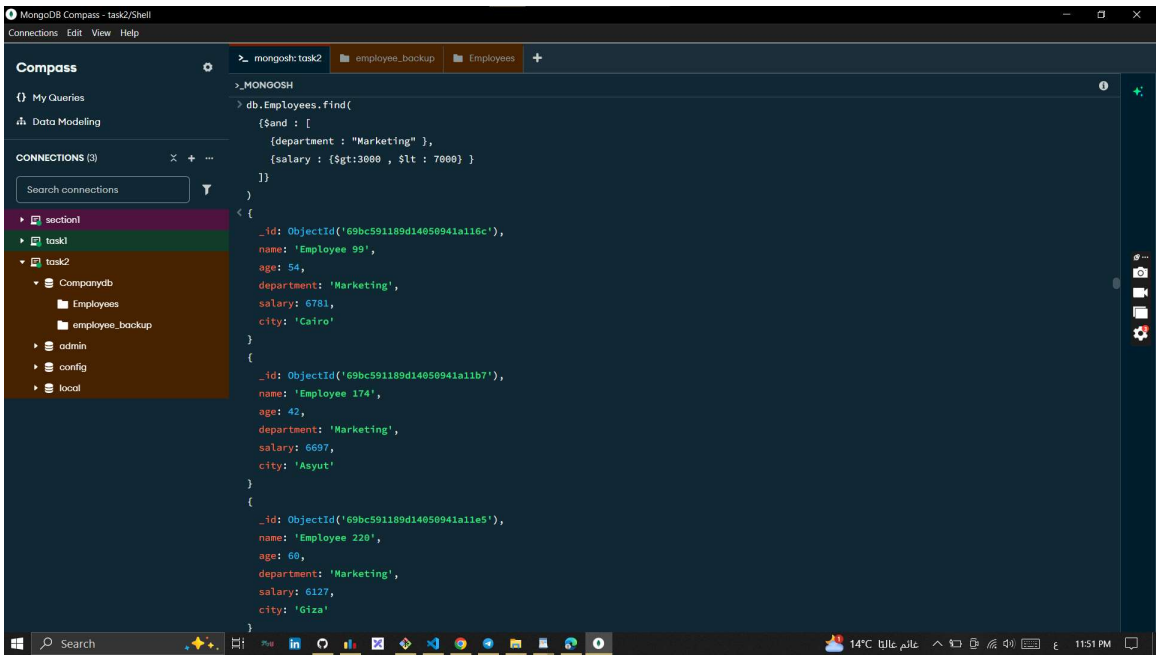
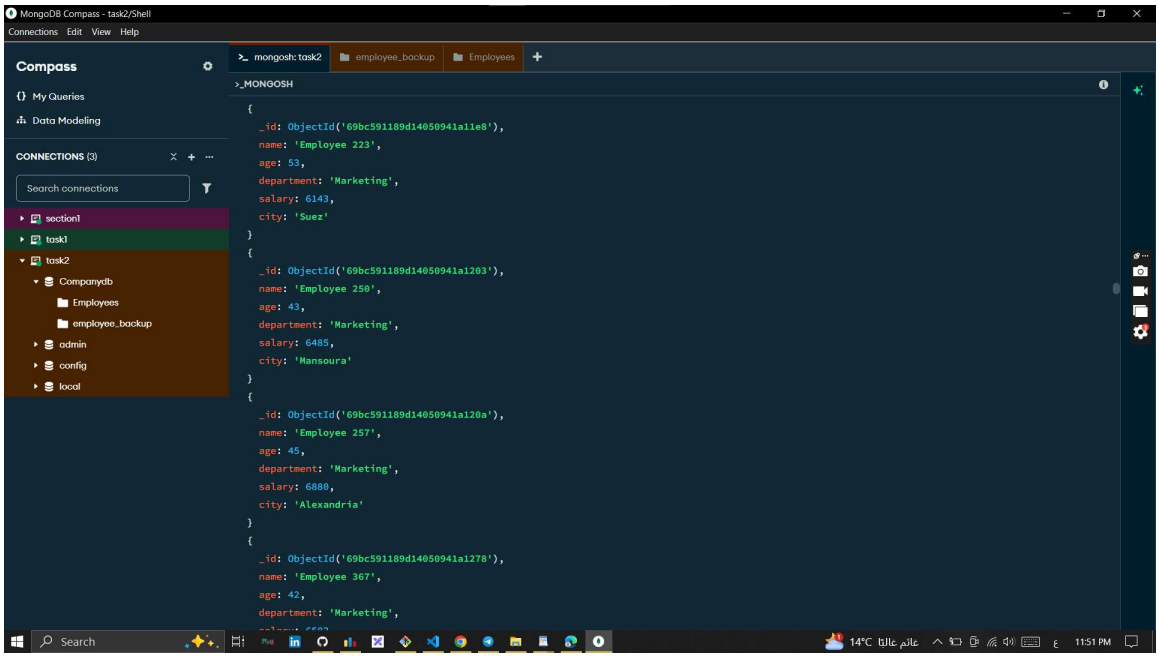


12) Get all employees working in "Marketing" and their salaries between 3000 and 7000.

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employee_backup
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>_MONGOSH
{
_id: ObjectId('69bc591189d14050941a11e8'),
name: 'Employee 223',
age: 53,
department: 'Marketing',
salary: 6143,
city: 'Suez'
}
{
_id: ObjectId('69bc591189d14050941a1203'),
name: 'Employee 250',
age: 43,
department: 'Marketing',
salary: 6485,
city: 'Mansoura'
}
{
_id: ObjectId('69bc591189d14050941a120a'),
name: 'Employee 257',
age: 45,
department: 'Marketing',
salary: 6880,
city: 'Alexandria'
}
{
_id: ObjectId('69bc591189d14050941a1278'),
name: 'Employee 367',
age: 42,
department: 'Marketing',
Search
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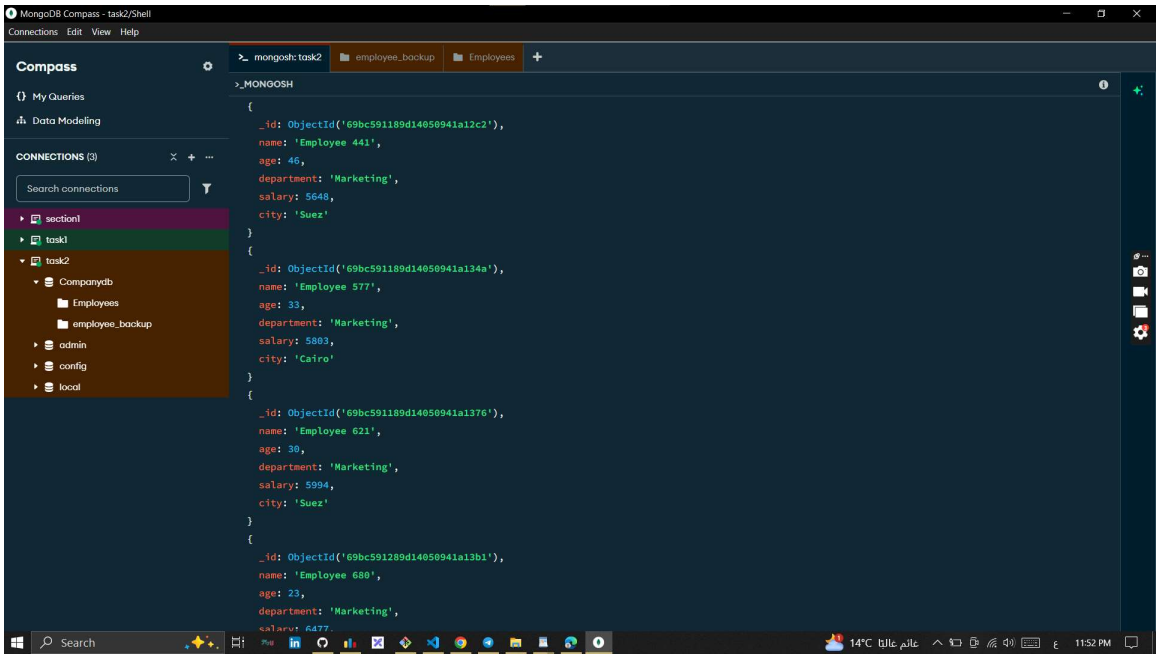
11:51 PM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task2
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employee_backup
admin
config
local
mongosh: task2
employee_backup
Employees
>_MONGOSH
{
_id: ObjectId('69bc591189d14050941a12c2'),
name: 'Employee 441',
age: 46,
department: 'Marketing',
salary: 5648,
city: 'Suez'
}
{
_id: ObjectId('69bc591189d14050941a134a'),
name: 'Employee 577',
age: 33,
department: 'Marketing',
salary: 5803,
city: 'Cairo'
}
{
_id: ObjectId('69bc591189d14050941a1376'),
name: 'Employee 621',
age: 30,
department: 'Marketing',
salary: 5994,
city: 'Suez'
}
{
_id: ObjectId('69bc591289d14050941a13b1'),
name: 'Employee 680',
age: 23,
department: 'Marketing',
Search
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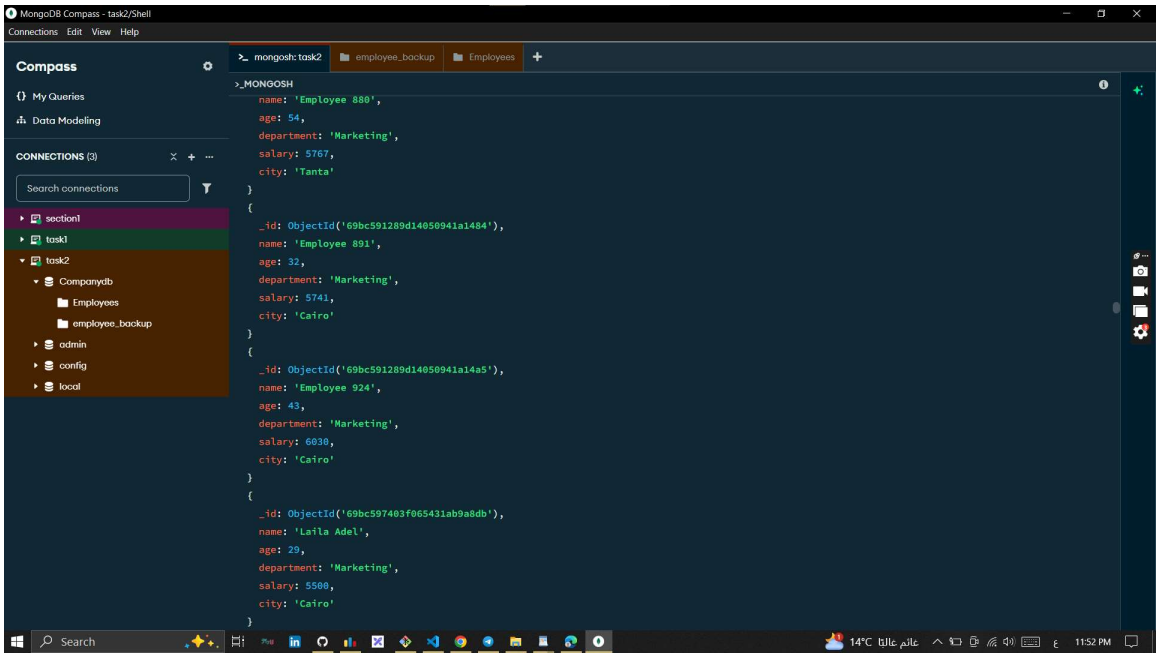
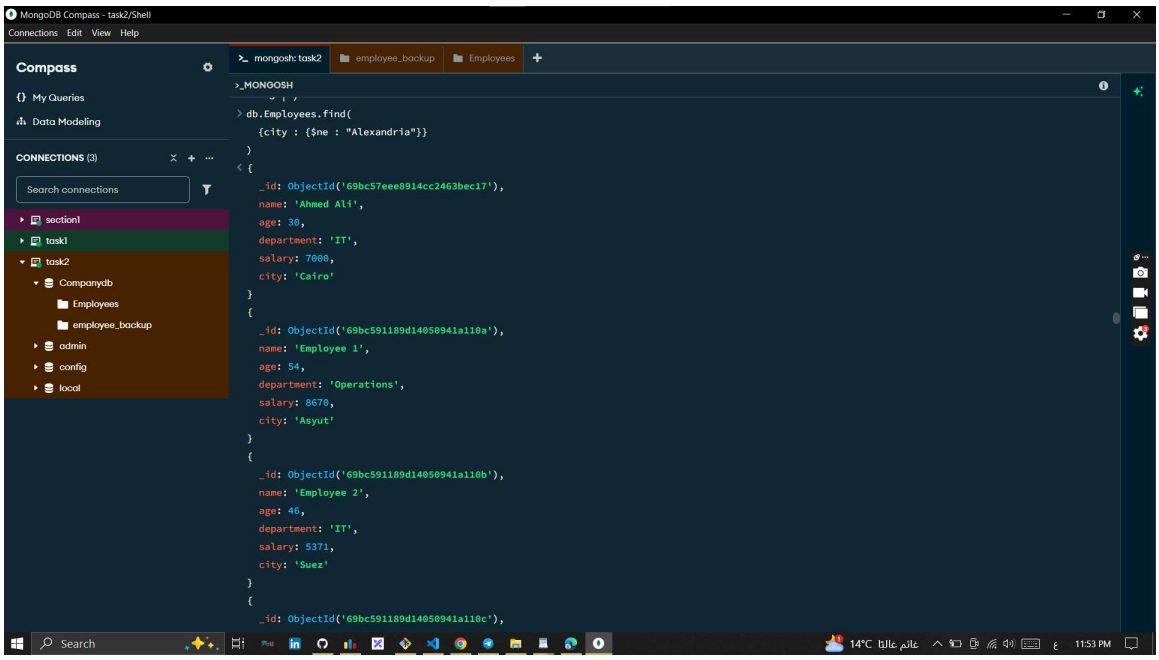
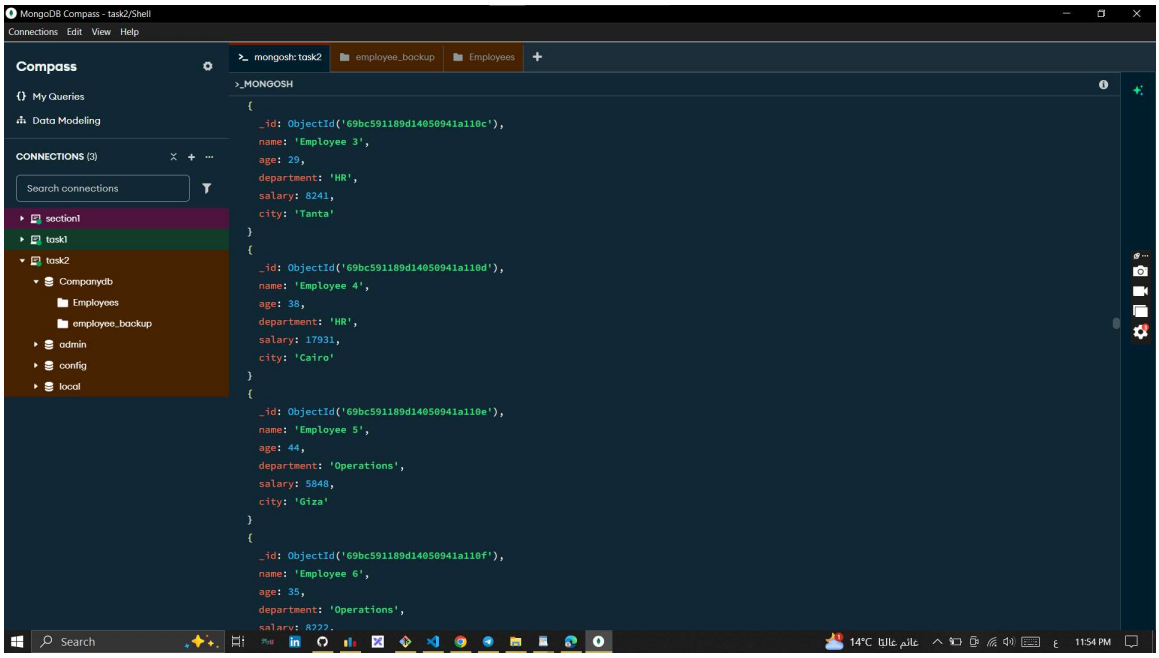


13) Look for employees who are not from the city of "Alexandria".

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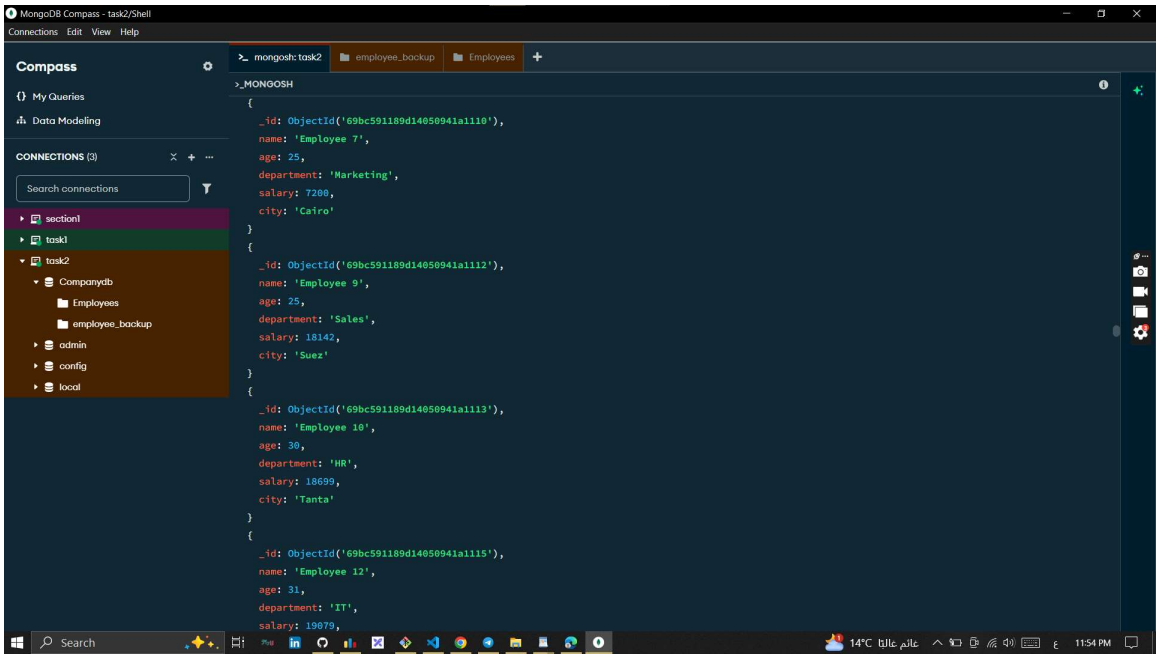
Employees
>_MONGOSH
{
_id: ObjectId('69bc591189d14050941a110c'),
name: 'Employee 3',
age: 29,
department: 'HR',
salary: 8241,
city: 'Tanta'
}
{
_id: ObjectId('69bc591189d14050941a110d'),
name: 'Employee 4',
age: 38,
department: 'HR',
salary: 17931,
city: 'Cairo'
}
{
_id: ObjectId('69bc591189d14050941a110e'),
name: 'Employee 5',
age: 44,
department: 'Operations',
salary: 5848,
city: 'Giza'
}
{
_id: ObjectId('69bc591189d14050941a110f'),
name: 'Employee 6',
age: 35,
department: 'Operations',
Search
14°C
11:54 PM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employee_backup
Employees
>_MONGOSH
{
_id: ObjectId('69bc591189d14050941a1110'),
name: 'Employee 7',
age: 25,
department: 'Marketing',
salary: 7200,
city: 'Cairo'
}
{
_id: ObjectId('69bc591189d14050941a1112'),
name: 'Employee 9',
age: 25,
department: 'Sales',
salary: 18142,
city: 'Suez'
}
{
_id: ObjectId('69bc591189d14050941a1113'),
name: 'Employee 10',
age: 30,
department: 'HR',
salary: 18699,
city: 'Tanta'
}
{
_id: ObjectId('69bc591189d14050941a1115'),
name: 'Employee 12',
age: 31,
department: 'IT',
salary: 19079,
Search
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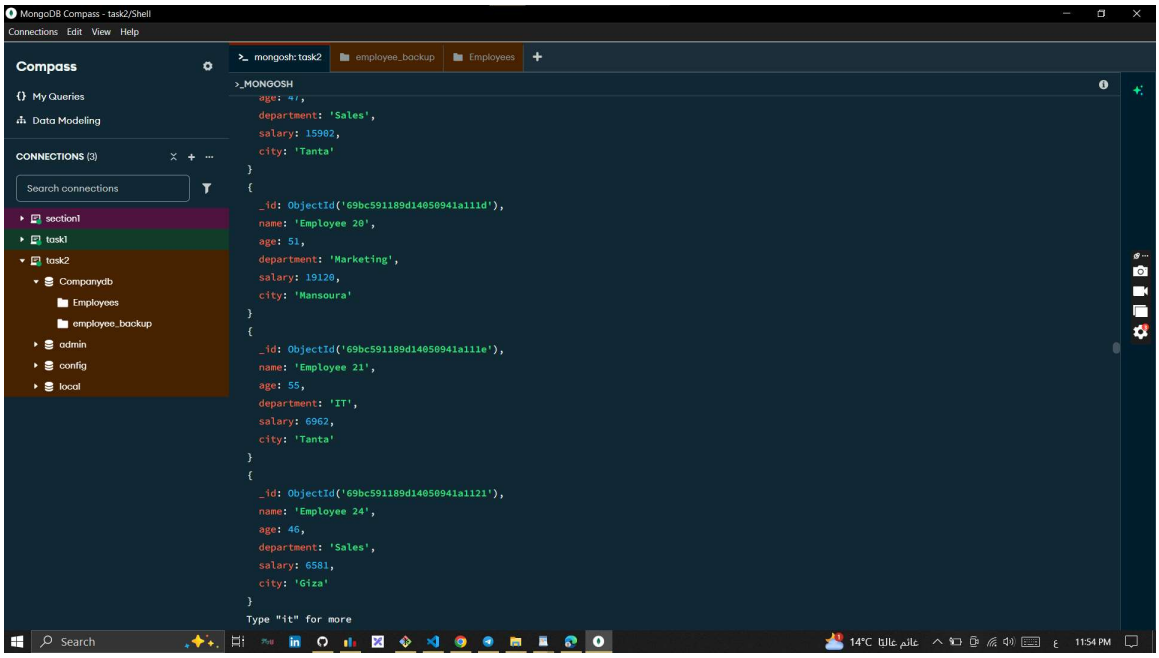
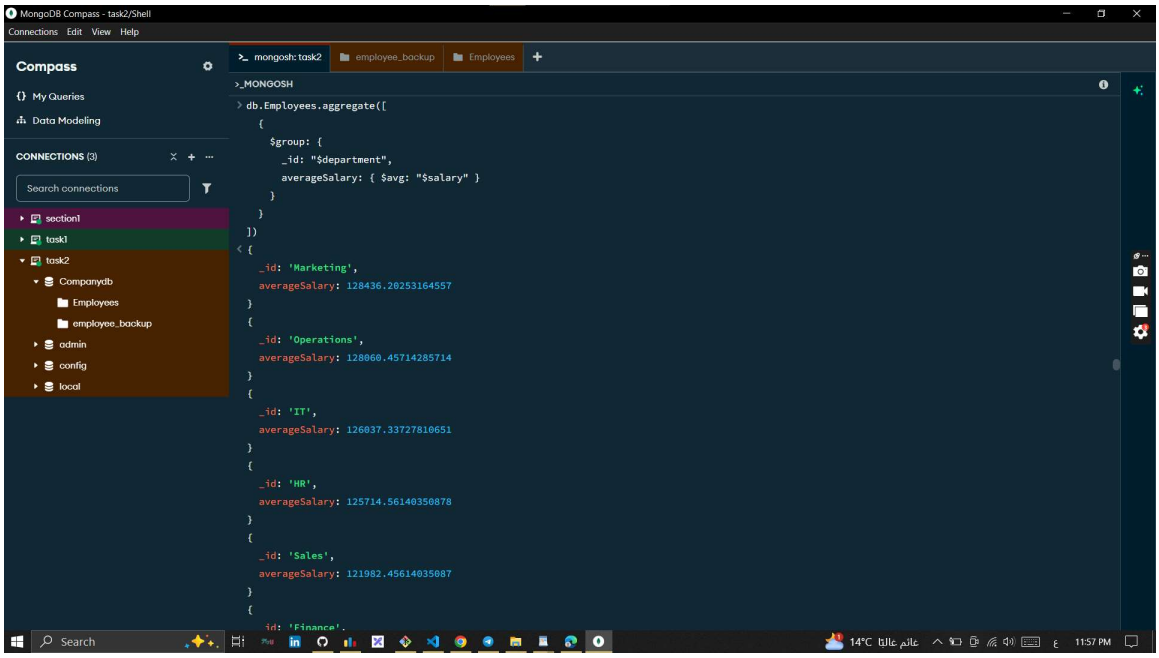


14) Update all employees in the Sales department and increase their salaries by 10%.

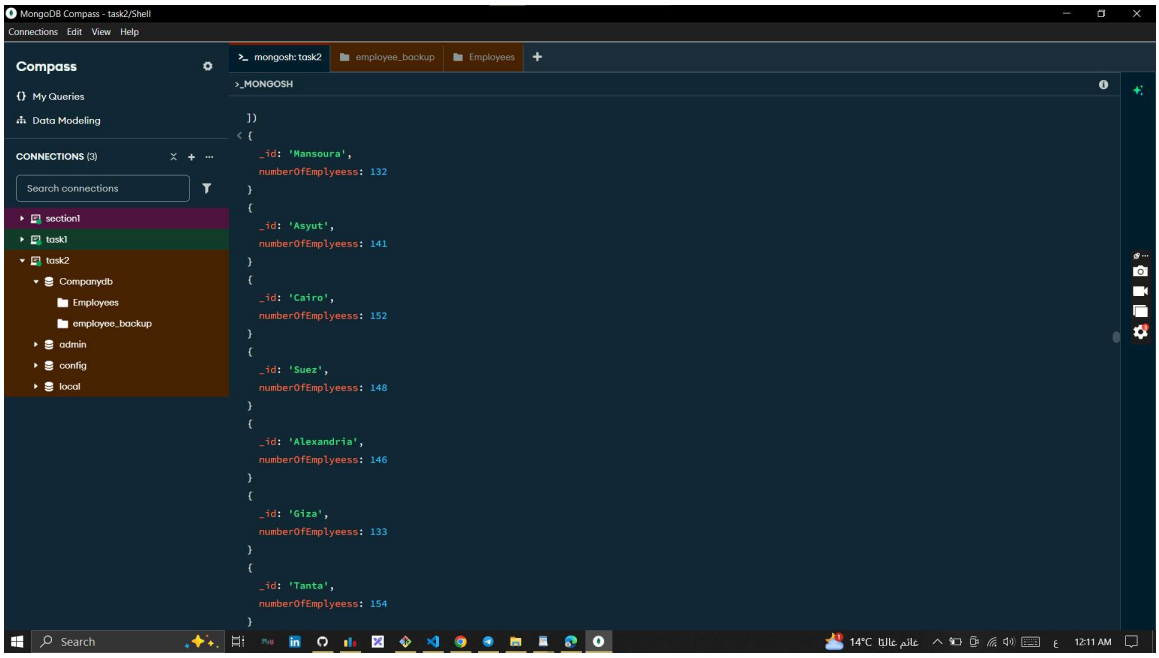
```
> db.Employees.updateMany( {},
                           {$mul : {salary : 10} })
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1006,
    modifiedCount: 1006,
    upsertedCount: 0
  }
```

15) Calculate the average salaries for each department.

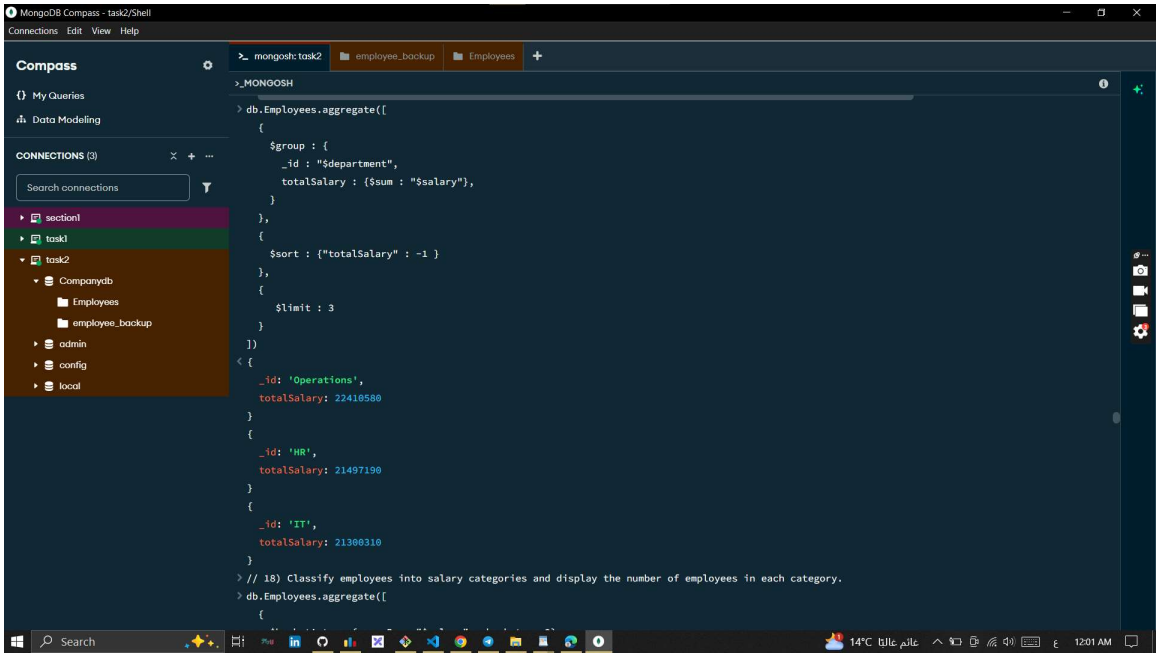


16) Display the number of employees in each city.

```
> db.Employees.aggregate([
    {
      $group : {
         _id : "$city",
         numberOfEmplyeess : {$sum : 1}
      }

    }



  ])
< {
    _id: 'Mansoura',
    numberOfEmplyeess: 132
  }
  {
    _id: 'Asyut',
    numberOfEmplyeess: 141
  }
  {
    _id: 'Cairo',
    numberOfEmplyeess: 152
  }
  {
    _id: 'Suez',
    numberOfEmplyeess: 148
  }
  {
    _id: 'Alexandria',
    numberOfEmplyeess: 146
```

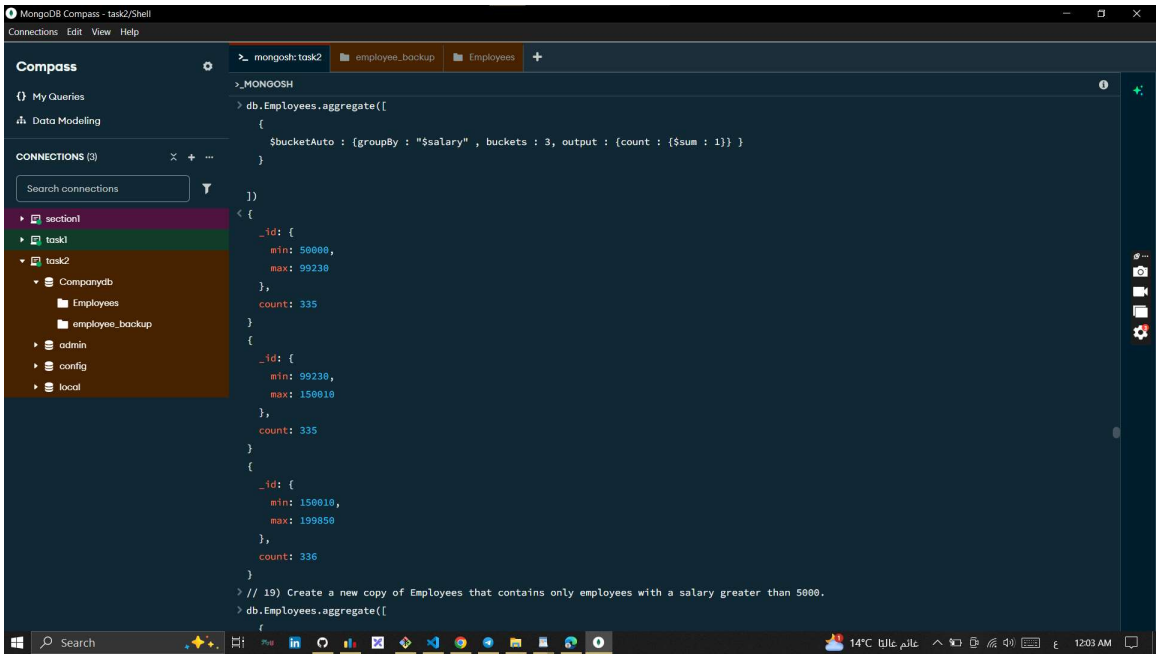


17) Find the top 3 departments by total salaries paid.



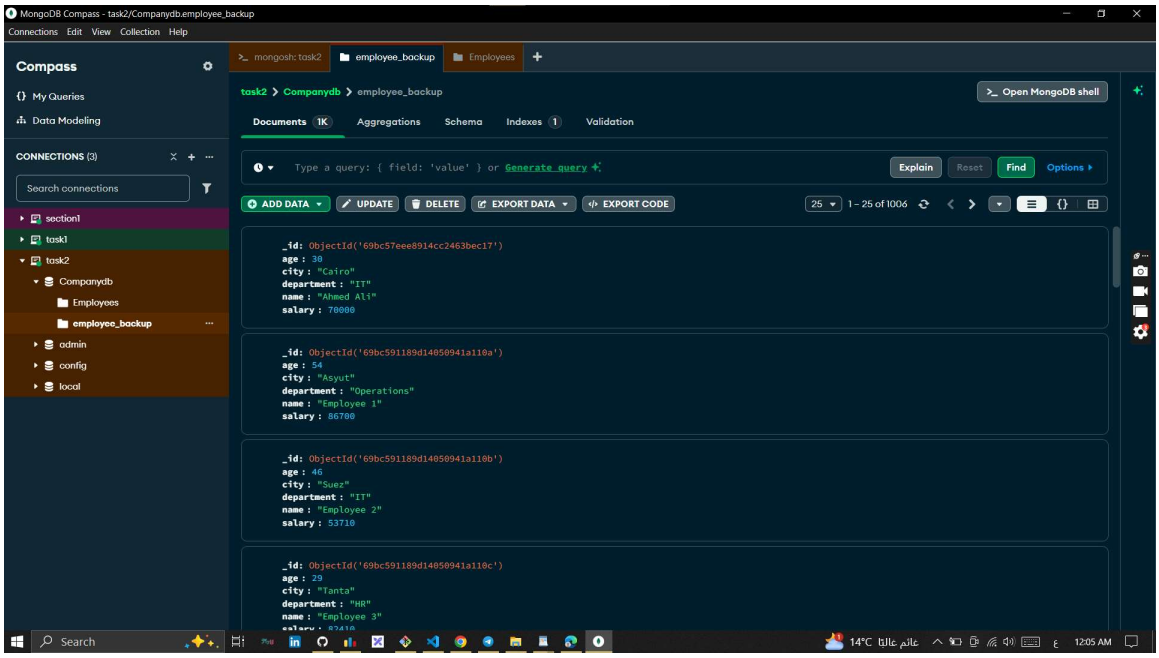
18) Classify employees into salary categories and display

the number of employees in each category.



19) Create a new copy of Employees that contains only employees with a salary greater than 5000.

```
> // 19) Create a new copy of Employees that contains only employees with a salary greater than 5000.
> db.Employees.aggregate([
    {
      $match : {salary : {$gt : 5000}}
    },
    {
      $merge : {into : "employee_backup" , whenMatched : "merge" , whenNotMatched : "insert"}
    }
  ])
<
> db.sales.insertMany{[
    { "employee_name": "Employee 1", "amount": 500 },
  { "employee_name": "Employee 5", "amount": 300 },
  { "employee_name": "Employee 7", "amount": 700 },
  { "employee_name": "Employee 12", "amount": 400 },
  { "employee_name": "Employee 20", "amount": 200 }
  ]}
```



20) Add a new sales collection like this:

{ "employee_name": "Employee 1", "amount": 500 },

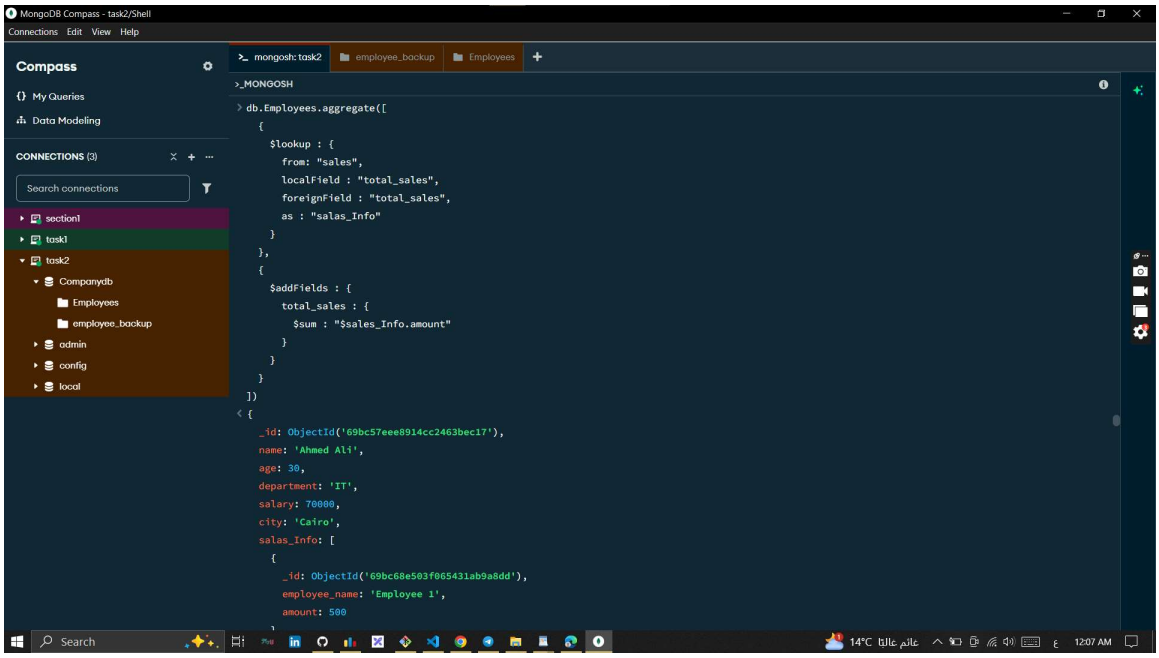
{ "employee_name": "Employee 5", "amount": 300 },

{ "employee_name": "Employee 7", "amount": 700 },

{ "employee_name": "Employee 12", "amount": 400 },

{ "employee_name": "Employee 20", "amount": 200 }

Then Calculate the total sales of each employee based on the sales recorded in Collection sales.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mongosh: task2
employee_backup
Employees
>_MONGOSH
db.Employees.aggregate([
{
$lookup : {
from: "sales",
localField : "total_sales",
foreignField : "total_sales",
as : "salas_Info"
}
},
{
$addFields : {
total_sales : {
$sum : "$sales_Info.amount"
}
}
}
])
{
_id: ObjectId('69bc57eee8914cc2463bec17'),
name: 'Ahmed Ali',
age: 30,
department: 'IT',
salary: 70000,
city: 'Cairo',
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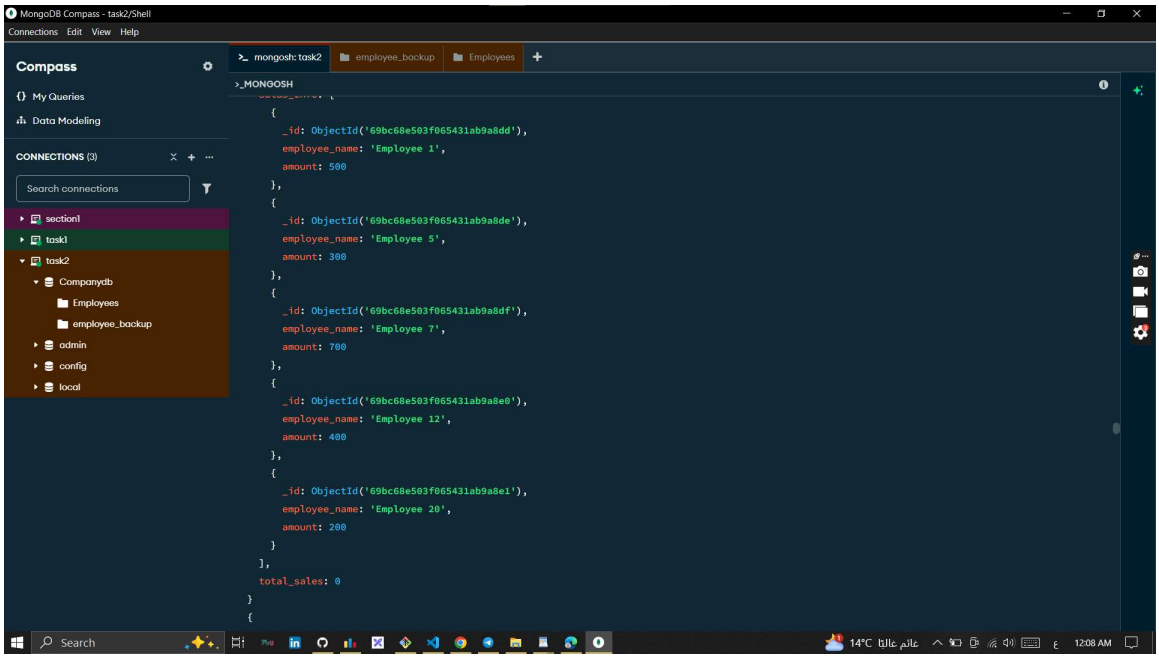
_id: ObjectId('69bc68e503f065431ab9a8dd'),
employee_name: 'Employee 1',
amount: 500
Search
14°C
12:07 AM


MongoDB Compass - task2/Shell
Connections Edit View Help
Compass
My Queries
Data Modeling
CONNECTIONS (3)
Search connections
section1
task1
task2
Companydb
Employees
employee_backup
admin
config
local
mongosh: task2
employee_backup
Employees
>_MONGOSH
{
_id: ObjectId('69bc68e503f065431ab9a8dd'),
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},
{
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employee_name: 'Employee 5',
amount: 300
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employee_name: 'Employee 7',
amount: 700
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amount: 200
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{
Search
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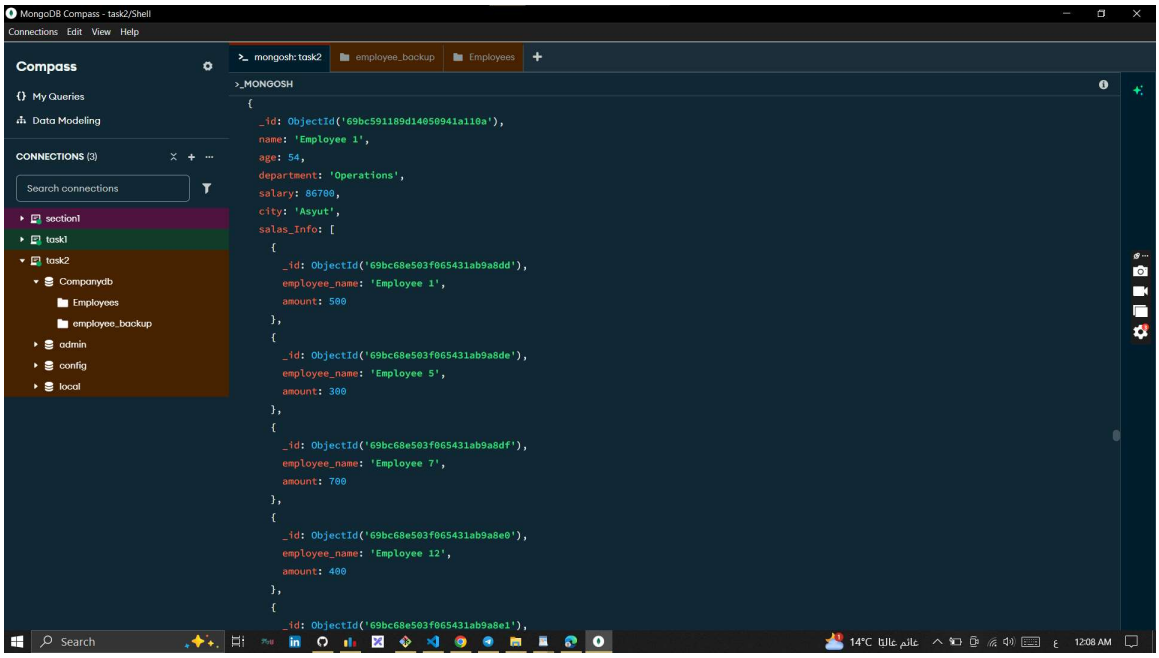
12:08 AM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config
local
mongosh: task2
employee_backup
Employees
>_MONGOSH
{
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name: 'Employee 1',
age: 54,
department: 'Operations',
salary: 86700,
city: 'Asyut',
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{
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employee_name: 'Employee 5',
amount: 300
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amount: 400
},
{
_id: ObjectId('69bc68e503f065431ab9a8e1'),
Search
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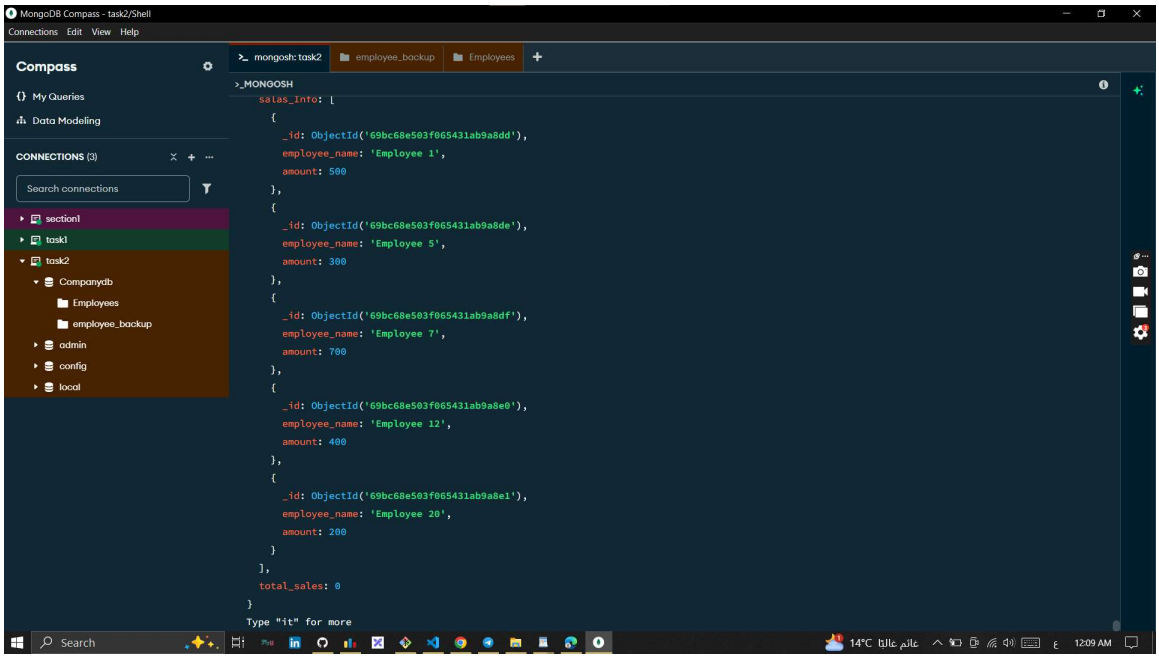
MongoDB Compass - task2/Shell
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amount: 400
},
{
_id: ObjectId('69bc68e503f065431ab9a8e1'),
employee_name: 'Employee 20',
amount: 200
}
],
total_sales: 0
}
Type "it" for more
Search
14°C
12:09 AM